BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

Objective Type

Time Allowed: 20 Min. | **Maximum Marks:** 17

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

Note: Objective Type

Q1. A ductile wire is stretched to double of its original length, %age elongation is ---:

(A) 200%
(B) 100%
(C) 400%
(D) 50%

Q2. Which is not true for ideal step up transformer:

(A) [illegible in source scan]
(B) [illegible in source scan]
(C) [illegible in source scan]
(D) [illegible in source scan]

Q3. A rod of length in a stationary frame is accelerated at speed of light. Its length measured perpendicular to its direction of motion is:

(A) [illegible in source scan]
(B) [illegible in source scan]
(C) [illegible in source scan]
(D) [illegible in source scan]

Q4. [question text blank in source scan]

(A) [illegible in source scan]
(B) [illegible in source scan]
(C) [illegible in source scan]
(D) [illegible in source scan]

Q5. Faraday and Maxwell unified electric and --- force :

(A) Strong nuclear
(B) Weak nuclear
(C) Gravitational
(D) Magnetic

Q6. The points where AC crosses the time axis corresponds to phase:

(A) [illegible in source scan]
(B) [illegible in source scan]
(C) [illegible in source scan]
(D) [illegible in source scan]

Q7. The potential of --- is least in CRO:

(A) Anode
(B) Grid
(C) Screen
(D) Cathode

Q8. The slope of graph between charge and time for capacitor charging is large initially when the product RC is :

(A) Small
(B) Large
(C) Intermediate
(D) Infinite

Q9. A combination of two back to back PN junctions is ---:

(A) Transistor
(B) Operational amplifier
(C) Digital gate
(D) Photo diode

Q10. If ionization energy of hydrogen atom is E, the energy required to remove electron from hydrogen in state n = 4 is:

(A) [illegible in source scan]
(B) [illegible in source scan]
(C) [illegible in source scan]
(D) [illegible in source scan]

Q11. The work on the principle of beats:

(A) DC motors
(B) Metal detectors
(C) Choke coils
(D) AC generators

Q12. A galvanometer coil of resistance Rg gives full scale deflection with current Ig. What is required shunt resistance Rs. if range of ammeter is I = 21g:

(A) Rg
(B) Rg/2
(C) 4 Rg
(D) 2 Rg

Q13. Energy required to remove all nucleons from nuclide of --- is maximum:

(A) [illegible in source scan]
(B) [illegible in source scan]
(C) [illegible in source scan]
(D) [illegible in source scan]

Q14. In alternating current ----- behave [illegible in source scan] resistors :

(A) Generators
(B) Inductors
(C) Capacitors
(D) Transformers

Q15. The fractional change in resistance is minimum for --- if temperature change is same for all :

(A) Platinum
(B) Constantan
(C) Nichrome
(D) Copper

Q16. The value of voltage gain of a transistor amplifier (common emitter) is of the order of:

(A) Millions
(B) Thousands
(C) Fraction
(D) Hundreds

Q17. Wave behaviour of matter is prominent at level:

(A) Macroscopic
(B) Microscopic
(C) Mega structure
(D) Any object size

Essay Type

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

> Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) Describe the force or forces on a positive charge when placed between parallel plates with opposite and equal charges.

(ii) If the distance between two point charges is halved, what will happen to the force between them?

(iii) What are the factors upon which the electric flux depend?

(iv) Why does capacitance of a parallel plate capacitor increase in the presence of a dielectric?

(v) At a given instant a proton moves in the positive x-direction in a region where there is a magnetic field in the negative z-direction. What is the direction of the magnetic force and direction of motion of proton?

(vi) How can a current loop be used to determine the presence of a magnetic field in a given region of space?

(vii) What is the importance of hair spring used in a Weston galvanometer? Explain.

(viii) Describe the working of an electron gun in CRO.

(ix) What is radiation tracer? Explain.

(x) Which radiation dose would deposit more energy to your body? (a) 10 mGy to your hand or (b) 1 mGy dose to your entire body.

(xi) How quenching is done in GM-tube?

(xii) How the scientists dispose off the radioactive waste safely?

Q3. (Marks 18) Write short answers to any EIGHT (8) questions:

(i) Why does the resistance of conductor rise with temperature?

(ii) A sinusoidal current has rms value of 10A. What is maximum or peak value?

(iii) What is meant by strain energy?

(iv) What is principle of virtual ground?

(v) Do bends in a wire affects its electrical resistance? Explain.

(vi) What is meant by A.M. and F.M.?

(vii) Define superconductor. Give example.

(viii) Why is the base current in a transistor is very small?

(ix) How rheo-state is used as potential divider?

(x) What is impedance? Give unit.

(xi) What is elastic limit of material in stress strain curve?

(xii) Give the application of gates in control system.

Q4. (Marks 12) Write short answers to any SIX (6) questions:

(i) Can a D.C motor be turned into DC generator? What changes are required be done?

(ii) In a transformer, there is no transfer of charge from the primary to the secondary. How is then the power transferred?

(iii) What is meant by armature?

(iv) Can pair production take place in vacuum? Explain.

(v) Will bright light eject more electrons from a metal surface dimmer light of same colour?

(vi) Is it possible to create a single electron from energy? Explain.

(vii) What are black body radiations? How can you get a black body?

(viii) How can the spectrum of hydrogen contain so many lines when hydrogen contains one electron?

(ix) Is energy conserved when an atom emits photon of light?

> Note: SECTION-II

> Note: Note: Attempt any THREE questions.

Q5.

(a) [5] Describe Millikan's oil drop experiment to determine charge on electron.

(b) A rectangular bar of iron is 2.0 cm by 2.0 cm in cross-section and 40 cm long. Calculate its resistance if the resistivity of iron is 11x10-8 ohm m. [printed marks position shows "?" - illegible in source scan]

Q6.

(a) [5] Derive the relation of of an electron.

(b) [3] An ideal step down transformer is connected to main supply of 240 V. It is desired to operate a 12 V, 30 W lamp. Find the current in the primary and the transformation ratio.

Q7.

(a) [5] What is RLC series circuit? Find out an expression for resonance frequency. Also write down its properties.

(b) [3] The current flowing into the base of a transistor is 100 micro A. Find its collector current and ratio, if the value of current gain beta is 100.

Q8.

(a) [5] What is hysteresis loop? Explain different terms, saturation, remanence and coercivity.

(b) [3] An electron is accelerated through a potential difference of 50 V. Calculate its de-Broglie wavelength.

Q9.

(a) [5] What is nuclear fission? Describe uncontrolled and controlled chain reaction.

(b) [3] Compute the shortest wavelength radiation in the Balmer Series. What value of n must be used?